

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

VECTOR CALCULUS V: The Divergence or Gauss theorem

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The next theorem relates a triple integral over a bounded region in $\mathbb{R}^3$ to a flux integral across the outer surface of the region.

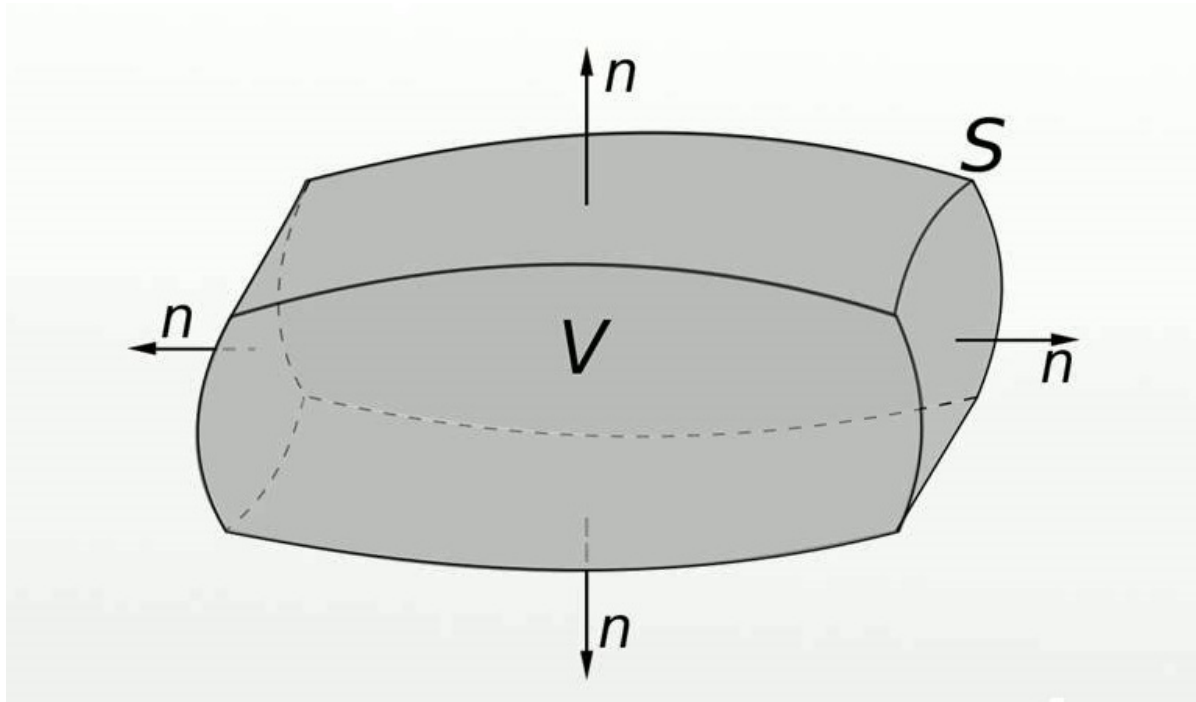


Figure 1: A closed piecewise regular surface $\mathcal{S}$, boundary of a region V . The picture is extracted from <https://en.wikipedia.org/wiki/Divergence-theorem>.

Theorem 0.1 (The Gauss Theorem or Divergence Theorem)

Let V be a closed, bounded region in $\mathbb{R}^3$ with a piecewise smooth boundary surface $\mathcal{S} = \partial V$, with an outward orientation. Let $\vec{F} = (P, Q, R) : U \rightarrow \mathbb{R}^3$ is a smooth, three-dimensional vector field defined in a region U containing the bounded region V and its boundary surface $\mathcal{S}$, then

$$\iint_{\mathcal{S}} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = \iiint_V \operatorname{div}(\vec{F})(x, y, z) dx dy dz . \quad (1)$$

If $\mathcal{S}$ is a closed surface, the evaluation of $\iint_{\mathcal{S}} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$ can often be simplified by applying the Gauss Theorem.

Example 0.2 Evaluate the flux integral $\iint_{\mathcal{S}} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$ where $\mathcal{S}$ is union of the hemisphere $x^2 + y^2 + (z - 1)^2 = 9$, $1 \leq z \leq 4$ and the plane $z = 1$ with $\vec{n}$ being an outward unit normal field and $\vec{F}(x, y, z) = (x, y, z - 1)$.

Solution: Try to evaluate

$$\iint_{\mathcal{S}} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = \iint_{\mathcal{S}_1} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS + \iint_{\mathcal{S}_2} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$$

directly, where $\mathcal{S}_1$ is the hemisphere $x^2 + y^2 + (z - 1)^2 = 9$, $1 \leq z \leq 4$ and $\mathcal{S}_2$ is the piece of plane $z = 1$ lying inside the sphere. Notice that $\mathcal{S}_2$ could also be described as the piece of the plane $z = 1$ lying inside the cylinder $x^2 + y^2 = 9$.

Notice that $\mathcal{S}_1$ is the graph of the function $g(x, y) = 1 + \sqrt{9 - x^2 - y^2}$ as defined on the disc $D := \{(x, y) \in \mathbb{R}^2: x^2 + y^2 \leq 9\}$ and use the identity (??) to evaluate $\iint_{\mathcal{S}_1} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$.

We get that

$$\begin{aligned} \iint_{\mathcal{S}_1} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS &= \iint_D \vec{F}(x, y, g(x, y)) \cdot (-g_x, -g_y, 1) dx dy \\ &= \iint_D (x, y, \sqrt{9 - x^2 - y^2}) \cdot \left(\frac{x}{\sqrt{9 - x^2 - y^2}}, \frac{y}{\sqrt{9 - x^2 - y^2}}, 1 \right) dx dy \\ &= \iint_D \left[\frac{x^2}{\sqrt{9 - x^2 - y^2}} + \frac{y^2}{\sqrt{9 - x^2 - y^2}} + \sqrt{9 - x^2 - y^2} \right] dx dy \\ &= \iint_D \frac{9}{\sqrt{9 - x^2 - y^2}} dx dy = \int_0^{2\pi} \int_0^3 \frac{9r}{\sqrt{9 - r^2}} dr d\theta \\ &= 18\pi \left[-\sqrt{9 - r^2} \right]_0^3 = 54\pi. \end{aligned}$$

Regarding the evaluation of $\iint_{\mathcal{S}_2} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$, notice that $\mathcal{S}_2$ lies on the plane $z = 1$ and that its unit normal vector field is $\vec{n}(x, y, z) = (0, 0, -1)$. (In fact, that surface is parametrized by the vector function $\vec{r}(s, t) = (s \cos t, s \sin t, 1)$ with $0 \leq s \leq 3$ and $0 \leq t \leq 2\pi$ for which $\vec{r}_s(s, t) \times \vec{r}_t(s, t) = (0, 0, -s)$) Hence,

$$\iint_{\mathcal{S}_2} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = \int_0^{2\pi} \int_0^3 (s \cos t, s \sin t, 0) \cdot (0, 0, -s) ds dt = 0.$$

This result was predictable since $\vec{n} = (0, 0, -1)$ and $\vec{F}(x, y, z) = (x, y, 0)$ on the surface $\mathcal{S}_2$. So,

$$\vec{F}(x, y, z) \cdot \vec{n}(x, y, z) = (x, y, 0) \cdot (0, 0, -1) = 0 \quad \text{on} \quad \mathcal{S}_2.$$

Therefore, the direct evaluation of $\iint_{\mathcal{S}} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$ gives

$$\boxed{\iint_{\mathcal{S}} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = \iint_{\mathcal{S}_1} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS + \iint_{\mathcal{S}_2} \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = 54\pi.}$$

On the other hand, using the divergence theorem (**having checked that the conditions for applying that theorem are satisfied**), we get

$$\iint_S \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = \iiint_V \operatorname{div}(\vec{F})(x, y, z) dx dy dz = \iiint_V 3 dx dy dz$$

where

$$V := \{(x, y, z) \in \mathbb{R}^3: x^2 + y^2 + (z - 1)^2 \leq 9, 1 \leq z \leq 4\}.$$

So, using the cylindrical coordinates, we find that

$$\begin{aligned} \iint_S \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS &= \iiint_V 3 dx dy dz = \int_0^{2\pi} \int_0^3 \int_1^{1+\sqrt{9-r^2}} 3r dz dr d\theta \\ &= 2\pi \int_0^3 3r\sqrt{9-r^2} dr = 2\pi \left[-(9-r^2)^{3/2} \right]_0^3 = 54\pi. \end{aligned}$$

As you can see the evaluation of the flux integral $\iint_S \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$ in this example is more tedious directly than using the divergence theorem.

Remark 0.3

- (i) There is an obvious similarity between the divergence theorem and both Green's theorem and Stokes' theorem. In fact, all the three theorems relate the integral of some combination of partial derivatives of a vector field (for instance $\operatorname{div}(\vec{F})$ for the divergence theorem) over a region to the integral of that vector field over the boundary of the region.
- (ii) It can be shown that both Stokes' and divergence theorems are reduced to the Green theorem for two dimensional vector fields and for surfaces lying entirely on the xy -plane. So,

GREEN THEO. = STOKES or DIV. THEO. in 2D

THE END OF THE COURSE!